

International Cybersecurity and Digital Forensics Academy (ICDFA)

GRC Engineering Programme

Programme Duration: 12 Months (52 Weeks)

Programme Mode: Online (Live Interactive Lectures, LMS, Hands-On Labs)

Credential Awarded: Certified GRC Engineer (CGRCE)

1.0 Programme Overview & Philosophy

The digital landscape is evolving at an unprecedented pace, moving beyond traditional check-box compliance. Modern organizations require professionals who can not only define governance policies but also engineer technical solutions to enforce them, measure their effectiveness, and automate compliance processes.

The **GRC Engineering Programme** is a cutting-edge, 12-month intensive course designed to bridge the critical gap between high-level governance theory and hands-on technical execution. This programme transforms participants into versatile **GRC Engineers** professionals capable of designing, building, and managing resilient, secure, and compliant systems.

Graduates of this programme will earn the **Certified GRC Engineer (CGRCE)** credential, signifying their expertise in implementing and automating Governance, Risk, and Compliance through technology.

2.0 Target Audience

This programme is ideal for:

- IT and Cybersecurity Professionals moving into GRC roles.
- Traditional GRC, Risk, and Compliance Analysts seeking to add technical skills.
- System and Network Administrators aiming to specialize in security and compliance.
- Security Auditors and IT Consultants.
- Graduates in Computer Science or Information Technology looking to build a specialized career.

3.0 Programme Learning Outcomes

Upon successful completion and certification, the CGRCE will be able to:

- Architect and implement a full GRC framework aligned with standards like NIST, ISO 27001, and PCI-DSS.

- Utilize Linux command-line tools and Python scripting to automate compliance checks and data aggregation.
- Integrate security controls into cloud environments (AWS, Azure, GCP) and DevOps pipelines.
- Configure and utilize security tools (SIEM, Vuln Scanners) for continuous monitoring and evidence collection.
- Design and execute technical incident response and business continuity plans.
- Translate regulatory requirements into enforceable technical policies and code.
- Communicate risk and compliance posture effectively to both technical and executive stakeholders.

4.0 Programme Curriculum Structure

The programme is divided into three intensive phases, culminating in a capstone project and the CGRCE certification.

Phase I: Compliance & Governance Foundations (18 Weeks)

This phase builds the foundational knowledge of GRC, enhanced with essential technical skills.

- **GRC101: Introduction to GRC and the Technical Landscape (4 Weeks)**
 - ✓ **Linux Fundamentals for GRC (Command Line, File Systems, Bash Scripting)**
 - ✓ Overview of GRC Frameworks and Principles
 - ✓ GRC Regulatory Environment and Standards
 - ✓ Risk Management Fundamentals
 - ✓ Compliance Management and Reporting
- **GRC102: Information Security Governance (5 Weeks)**
 - ✓ Principles of Information Security Governance
 - ✓ Developing Security Policies and Procedures
 - ✓ Roles and Responsibilities in Governance
 - ✓ Monitoring and Auditing Security Controls

- ✓ Case Studies in Security Governance

- **GRC103: Risk Assessment and Management Techniques (5 Weeks)**

- ✓ Risk Identification Techniques
- ✓ Risk Analysis and Evaluation
- ✓ Risk Treatment Strategies
- ✓ Risk Monitoring and Reporting
- ✓ Practical Risk Assessment Workshop

- **GRC104: Compliance Frameworks and Legal Requirements (4 Weeks)**

- ✓ Overview of Global Compliance Frameworks (NIST, ISO 27001, CIS, PCI-DSS)
- ✓ Data Privacy Laws and Regulations (GDPR, CCPA, NDPA)
- ✓ Legal Implications of Non-Compliance
- ✓ Compliance Audit Processes

- **Phase 1 Competency Examination & Mid-Term Break**

Phase II: Advanced GRC & Technical Integration (17 Weeks)

This phase focuses on applying GRC principles to advanced scenarios and integrating with core security operations.

- **GRC201: Technical Incident Management and Business Continuity (5 Weeks)**

- ✓ Incident Response Planning and Playbook Development
- ✓ Digital Forensics Fundamentals for GRC
- ✓ Business Continuity and Disaster Recovery Planning
- ✓ Crisis Communication and Stakeholder Management
- ✓ Tabletop Incident Handling Exercises

- **GRC202: Security Awareness Engineering (3 Weeks)**

- ✓ Designing and Implementing Security Awareness Programs

- ✓ Behavioral Psychology and Risk Reduction
- ✓ Phishing Simulation Campaigns & Evaluating Training Effectiveness
- **GRC203: Cloud GRC and Automation (5 Weeks)**
 - ✓ GRC for Cloud Environments (AWS, Azure, GCP)
 - ✓ Cybersecurity Risk in Cloud and IoT Environments
 - ✓ **Introduction to Python for GRC Automation**
 - ✓ Automating Compliance Checks with Scripts
 - ✓ Regulatory Changes and Future Compliance Challenges
- **December Break**
- **GRC204: Security Operations for GRC Engineers (4 Weeks)**
 - ✓ Introduction to SIEM (Splunk, Elastic) for Compliance Logging
 - ✓ Vulnerability Management Lifecycle
 - ✓ Configuration Management and Hardening (CIS Benchmarks)
 - ✓ Interpreting Penetration Test Reports for Risk Management
- **Phase 2 Competency Examination**

Phase III: GRC Implementation & Automation (12 Weeks)

This is the engineering core, where students learn to build and automate GRC solutions.

- **GRC301: DevSecOps and Infrastructure as Code (IaC) for GRC (4 Weeks)**
 - ✓ Introduction to DevSecOps Principles
 - ✓ Secure SDLC and Policy as Code (Open Policy Agent)
 - ✓ IaC Security with Terraform
 - ✓ Continuous Compliance Monitoring in CI/CD Pipelines
- **GRC302: GRC Tooling and Platform Engineering (4 Weeks)**
 - ✓ Survey of GRC Platforms (e.g., ServiceNow, RSA Archer)

- ✓ Designing a Unified Compliance Dashboard
- ✓ API Integration for GRC Data Aggregation
- ✓ Building Custom Workflows for Risk Exception Handling
- **GRC399: Capstone Project - Engineering a GRC Program (4 Weeks)**
 - ✓ Capstone Project Kick-off and Scope Finalization
 - ✓ Capstone Development & Mentorship Sessions
 - ✓ Capstone Report and Solution Submission
 - ✓ **Final Presentations and Graduation Ceremony**

5.0 Capstone Project Focus Areas

Participants will select a real-world project, such as:

- Developing and automating a cloud compliance check system for PCI-DSS.
- Building an integrated risk dashboard by aggregating data from SIEM, Vulnerability Scanner, and Cloud APIs.
- Creating a set of security policies as code for a Kubernetes deployment using IaC.
- Engineering a full incident response automation runbook with playbooks.
- Designing and documenting a technical control framework for a specific regulation (e.g., NDPA).

6.0 Assessment and CGRCE Certification Award

To be awarded the **Certified GRC Engineer (CGRCE)** credential, participants must complete all of the following:

1. **Continuous Assessment (40% of Final Grade)**
 - ✓ Weekly assignments and quizzes.
 - ✓ Hands-on technical lab exercises and reports.
 - ✓ Participation in live sessions and workshops.
2. **Phase 1 & 2 Competency Examinations (30% of Final Grade)**
 - ✓ **Paper 1 (Objective):** Multiple-choice and short-answer questions.

- ✓ **Paper 2 (Practical):** Hands-on lab-based challenges.
- ✓ **Paper 3 (Oral Professional Discussion):** Scenario-based viva voce.

3. Capstone Project (30% of Final Grade)

- ✓ **Project Submission:** Complete report, code, and documentation.
- ✓ **Final Defense:** Presentation and Q&A before an industry panel.

A minimum cumulative score of 70% is required to pass the programme and be eligible for the CGRCE certification.

8.0 Admission Requirements

- A bachelor's degree in Computer Science, Information Technology, or a related field. Significant relevant work experience may be considered in lieu of a degree.
- Foundational knowledge of networking and information security is recommended.
- Applicants must complete an application form and may be required to participate in an admissions interview.

Become a Pioneer in the Future of GRC. Engineer Security and Compliance.

For further inquiries and applications, contact the ICDFA Training Office:

Email: training@icdfa.edu.ng

Official Portal: www.icdfa.edu.ng